

A Technical Introduction to Semi-supervised learning

Section 1

An exact solution is intractable due to the non-convex term $(1 - |f(x)|)_{+}$, so research focuses on useful approximations. Other approaches that implement low-density separation include Gaussian process models, information regularization, and entropy minimization (of which TSVM is a special case).

Section 2

Assumptions In order to make any use of unlabeled data, some relationship to the underlying distribution of data must exist. Semi-supervised learning algorithms make use of at least one of the following assumptions:

Section 3

$$\mathbf{f}^T \mathbf{L} \mathbf{f} = \sum_{i,j=1}^{l+u} W_{ij} (f_i - f_j)^2 \approx \int \mathbf{M} \cdot \nabla \mathbf{M} f(x) \cdot \nabla \mathbf{p}(x) \quad \{\displaystyle \mathbf{f}^T \mathbf{L} \mathbf{f} = \sum_{i,j=1}^{l+u} W_{ij} (f_i - f_j)^2 \approx \int \mathbf{M} \cdot \nabla \mathbf{M} f(x) \cdot \nabla \mathbf{p}(x)\}$$

The graph-based approach to Laplacian regularization is to put in relation with finite difference method. The Laplacian can also be used to extend the supervised learning algorithms: regularized least squares and support vector machines (SVM) to semi-supervised versions Laplacian regularized least squares and Laplacian SVM.

Section 4

$$\argmin_{f \in H} \left(\frac{1}{l} \sum_{i=1}^l V(f(x_i), y_i) + \lambda A \cdot \mathbf{f}^T \mathbf{H}^{-1} \mathbf{f} + \lambda \int \mathbf{M} \cdot \nabla \mathbf{M} f(x) \cdot \nabla \mathbf{p}(x) \right) \quad \{\displaystyle \argmin_{f \in H} \left(\frac{1}{l} \sum_{i=1}^l V(f(x_i), y_i) + \lambda A \cdot \mathbf{f}^T \mathbf{H}^{-1} \mathbf{f} + \lambda \int \mathbf{M} \cdot \nabla \mathbf{M} f(x) \cdot \nabla \mathbf{p}(x) \right)\}$$

Section 5

More formally, semi-supervised learning assumes a set of l independently identically distributed examples $x_1, \dots, x_l \in X$ with corresponding labels $y_1, \dots, y_l \in Y$ and u unlabeled examples $x_{l+1}, \dots, x_{l+u} \in X$ are processed. Semi-supervised learning combines this information to surpass the classification performance that can be obtained either by discarding the unlabeled data and doing supervised learning or by discarding the labels and doing unsupervised learning. Semi-supervised learning may refer to either transductive learning or inductive learning. The goal of transductive learning is to infer the correct labels for the given unlabeled data $x_{l+1}, \dots, x_{l+u}$.

only. The goal of inductive learning is to infer the correct mapping from X to Y . It is unnecessary (and, according to Vapnik's principle, imprudent) to perform transductive learning by way of inferring a classification rule over the entire input space; however, in practice, algorithms formally designed for transduction or induction are often used interchangeably.

Section 6

Heuristic approaches Some methods for semi-supervised learning are not intrinsically geared to learning from both unlabeled and labeled data, but instead make use of unlabeled data within a supervised learning framework. For instance, the labeled and unlabeled examples $x_1, \dots, x_{l+u}$ may inform a choice of representation, distance metric, or kernel for the data in an unsupervised first step. Then supervised learning proceeds from only the labeled examples. In this vein, some methods learn a low-dimensional representation using the supervised data and then apply either low-density separation or graph-based methods to the learned representation. Iteratively refining the representation and then performing semi-supervised learning on said representation may further improve performance. Self-training is a wrapper method for semi-supervised learning. First a supervised learning algorithm is trained based on the labeled data only. This classifier is then applied to the unlabeled data to generate more labeled examples as input for the supervised learning algorithm. Generally only the labels the classifier is most confident in are added at each step. In natural language processing, a common self-training algorithm is the Yarowsky algorithm for problems like word sense disambiguation, accent restoration, and spelling correction. Co-training is an extension of self-training in which multiple classifiers are trained on different (ideally disjoint) sets of features and generate labeled examples for one another.

Section 7

In human cognition Human responses to formal semi-supervised learning problems have yielded varying conclusions about the degree of influence of the unlabeled data. More natural learning problems may also be viewed as instances of semi-supervised learning. Much of human concept learning involves a small amount of direct instruction (e.g. parental labeling of objects during childhood) combined with large amounts of unlabeled experience (e.g. observation of objects without naming or counting them, or at least without feedback). Human infants are sensitive to the structure of unlabeled natural categories such as images of dogs and cats or male and female faces. Infants and children take into account not only unlabeled examples, but the sampling process from which labeled examples arise.

Section 8

The acquisition of labeled data for a learning problem often requires a skilled human agent (e.g. to transcribe an audio segment) or a physical experiment (e.g. determining the 3D structure of a protein or determining whether there is oil at a particular location). The cost associated with the labeling process thus may render large, fully labeled training sets infeasible, whereas acquisition of unlabeled data is relatively inexpensive. In such situations, semi-supervised learning can be of great practical value. Semi-supervised learning is also of theoretical interest in machine learning and as a model for human learning.

Section 9

History The heuristic approach of self-training (also known as self-learning or self-labeling) is historically the oldest approach to semi-supervised learning, with examples of applications starting in the 1960s. The transductive learning framework was formally introduced by Vladimir Vapnik in the 1970s. Interest in inductive learning using generative models also began in the 1970s. A probably approximately correct learning bound for semi-supervised learning of a Gaussian mixture was demonstrated by Ratsaby and Venkatesh in 1995.

Section 10

Generative models Generative approaches to statistical learning first seek to estimate $p(x|y)$, the distribution of data points belonging to each class. The probability $p(y|x)$ that a given point x has label y is then proportional to $p(x|y)p(y)$ by Bayes' rule. Semi-supervised learning with generative models can be viewed either as an extension of supervised learning (classification plus information about $p(x)$) or as an extension of unsupervised learning (clustering plus some labels). Generative models assume that the distributions take some particular form $p(x|y, \theta)$ parameterized by the vector θ . If these assumptions are incorrect, the unlabeled data may actually decrease the accuracy of the solution relative to what would have been obtained from labeled data alone. However, if the assumptions are correct, then the unlabeled data necessarily improves performance. The unlabeled data are distributed according to a mixture of individual-class distributions. In order to learn the mixture distribution from the unlabeled data, it must be identifiable, that is, different parameters must yield different summed distributions. Gaussian mixture distributions are identifiable and commonly used for generative models. The parameterized joint distribution can be written as $p(x, y | \theta) = p(y | \theta) p(x | y, \theta)$ by using the chain rule. Each parameter vector θ is associated with a decision function $f_{\theta}(x) = \underset{y}{\operatorname{argmax}} p(y | x, \theta)$

$p(y|x, \theta)$. The parameter is then chosen based on fit to both the labeled and unlabeled data, weighted by λ :